# Pulsar Identification and Categorization

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We worked on using machine learning to identify real pulsar signals among radio observations and explore similarities within the confirmed pulsars. The project has two parts: supervised classification for separating pulsars from noise, and unsupervised learning for finding groups of similar signals.

**Data preparation**
We used the HTRU2 dataset, which contains 17,898 candidates described by eight numerical features. Of these, 1,639 are real pulsars and 16,259 are noise or interference. We split the data into 70% training, 15% validation and 15% testing while keeping the class proportions similar. We also extracted the confirmed pulsars into a separate file for clustering.

**Identifying pulsars**
We implemented XGBoost, Random Forest and an RBF Support Vector Machine. The notebooks handle the class imbalance through class weights, tune model settings and choose decision thresholds using validation data. The saved test results show that Random Forest and SVM gave similar performance. With tuned thresholds, Random Forest achieved 87.1% precision and 87.8% recall, while SVM achieved 85.0% precision and 89.8% recall. XGBoost reached 92.7% recall, but precision fell to 45.1%, showing the cost of catching more pulsars through extra false alarms.

**Exploring pulsar groups**
We applied K-Means to the 1,639 confirmed pulsars using the original feature values. With three clusters, it produced groups of 974, 621 and 44 signals, with a silhouette score of 0.4413. We also built a 5 by 5 Self-Organizing Map from scratch using NumPy. After mapping its neurons to K-Means centroids, it produced two final groups, with a silhouette score of 0.4024. PCA reduced the eight features to two components that explain 86.04% of the variation and show a boomerang-shaped pattern.

**Presenting the work**
We created plots, clustered data files, saved classification results and detailed reports. We also built a Streamlit dashboard with interactive plots, feature comparisons and a sandbox for exploring cluster assignments.

**Current status and conclusion**
Overall, the models were highly successful in both detecting genuine pulsar emissions and discovering structural patterns within confirmed signals. The supervised classifiers effectively handled extreme class imbalance, with Random Forest and SVM delivering balanced precision and recall near 90%, and XGBoost maximizing candidate recovery at 92.7% recall. Additionally, K-Means and the custom Self-Organizing Map proved successful in partitioning confirmed signals into distinct, well-separated clusters with strong silhouette metrics and clear PCA representations. This project demonstrates a robust, end-to-end framework capable of accelerating real-world radio pulsar verification.